

COMS 4720/5720: Homework 7
Due: Apr 3th, 11:59pm
Total Points: 40

Late submission policy. An assignment that is submitted one day late will get a penalty of 10%. An assignment that is submitted two days late will get a penalty of 20%. Assignments submitted after two days will not be graded and will get no points. For example, if an assignment is due on Friday by midnight, a submission on Saturday will be penalized by 10%, and a submission on Sunday will be penalized by 20%. A submission on Monday will get no points.

Submission format. Homework solutions must be **handwritten**. You may either write directly on a tablet or write on paper and scan your work. In both cases, the final submission must be compiled into a **single PDF file**. Typed solutions are not permitted unless you obtain prior, explicit permission from the instructor. We reserve the right **NOT** to grade homework that does not follow these requirements.

The submission file must be named as follows:

- <Your-net-id>-4720-hw7.pdf for COMS 4720
- <Your-net-id>-5720-hw7.pdf for COMS 5720.

For instance, if your netid is **asterix** and you are enrolled COMS 4720, then your submission file should be named **asterix-4720-hw7.pdf**. Each student must hand in their own assignment. If you discussed the homework or solutions with others, a list of collaborators must be included with each submission. Each of the collaborators has to write the solutions in their own words (copies are not allowed). Also, recall the AI policy:

- You may **not use AI tools to generate complete solutions** to assignments, coding problems, or any assessed work that is intended to measure your independent knowledge and skills.
- Whenever you use AI tools to assist in your assignment, you must **include a paragraph at the end of the assignment explaining what you used the AI for and include all the prompts you used to get the results**. Failure to do so is in violation of the academic honesty policies.

1. **(20 pts)** Solve the cryptarithmic problem shown in Figure 6.2 of the textbook by hand using backtracking search with forward checking. Apply the Minimum Remaining Values (MRV) heuristic for variable selection and the Least Constraining Value (LCV) heuristic for value selection.

Perform the search step by step. At each step, explicitly state which variable is selected and justify the choice using the MRV heuristic. For the selected variable, list its current domain and evaluate possible values using the LCV heuristic, explaining why the chosen value is preferred. After each assignment, perform forward checking and clearly indicate how the domains of remaining variables are updated. Whenever a contradiction arises, identify precisely which constraint is violated, indicate the point to which the algorithm backtracks, and continue the search systematically until a complete consistent assignment is found.

Finally, provide the complete assignment that satisfies all constraints. Solutions that present only the final assignment without showing the detailed search process will not receive credit.

2. (20 pts) Consider the following logic puzzle. In five houses, each with a different color, live five persons of different nationalities, each of whom prefers a different brand of candy, a different drink, and a different pet. Given the following fourteen facts, the questions to answer are “Where does the zebra live, and in which house do they drink water?”

- (i) The Englishman lives in the red house.
- (ii) The Spaniard owns the dog.
- (iii) The Norwegian lives in the first house on the left.
- (iv) The green house is immediately to the right of the ivory house.
- (v) The man who eats Hershey bars lives in the house next to the man with the fox.
- (vi) Kit Kats are eaten in the yellow house.
- (vii) The Norwegian lives next to the blue house.
- (viii) The Smarties eater owns snails.
- (ix) The Snickers eater drinks orange juice.
- (x) The Ukrainian drinks tea.
- (xi) The Japanese eats Milky Ways.
- (xii) Kit Kats are eaten in a house next to the house where the horse is kept.
- (xiii) Coffee is drunk in the green house.
- (xiv) Milk is drunk in the middle house.

Assume that the five houses are numbered 1, 2, 3, 4, and 5 from left to right. Formulate a **complete and mathematically precise** CSP representation of the problem by answering the following questions.

- (a) **(8 pts)** Mathematically and precisely define all variables of your CSP model. For each variable, clearly state what it represents and specify its domain using explicit set notation. Your answer must consist of a mathematically well-defined set of variables together with a domain specification for each variable. Answers that list informal descriptions without explicit domains will be considered incomplete.

- (b) **(2 pts)** State the total number of variables in your model and justify your answer based on your definition in part (a). The number must follow directly from your formal specification of the variable set.
- (c) **(7 pts)** Translate each of the 14 clues into formal constraints over the variables defined in part (a). Each clue must be expressed as a logical or arithmetic constraint using precise mathematical notation (e.g., equality, implication, quantifiers, etc.). Informal English restatements of the clues will not receive credit.

- (d) **(3 pts)** Formally specify all structural constraints required by your modeling assumptions. In particular, you must explicitly state any all-different constraints or uniqueness conditions. These constraints must be written mathematically (e.g., using AllDiff or an equivalent formal definition).